DATABASE

Q1. A database is best defined as:
A) A collection of programs
B) A collection of related data stored in an organized manner
C) A spreadsheet system
D) A programming language
ANSWER: B

Q2. A Database Management System (DBMS) is:
A) A collection of SQL statements
B) Software used to manage databases
C) A hardware device
D) A compiler
ANSWER: B

Q3. Which of the following is an advantage of DBMS?
A) Data redundancy
B) Data inconsistency
C) Data independence
D) Data duplication
ANSWER: C

Q4. The person responsible for managing database security and access is:
A) Programmer
B) Database Administrator
C) End user
D) System analyst
ANSWER: B

Q5. Data stored in a database is called:
A) Metadata
B) Raw data
C) Structured data
D) Information
ANSWER: C

Q6. The three-schema architecture was proposed to achieve:
A) Data independence
B) Data redundancy
C) Data duplication
D) Data fragmentation
ANSWER: A

Q7. The external level of database architecture describes:
A) Physical storage
B) User view of data
C) Hardware details
D) File allocation
ANSWER: B

Q8. The internal schema describes:
A) Physical storage structure
B) User interfaces
C) Application logic
D) Data queries
ANSWER: A

Q9. Logical data independence refers to:
A) Changing physical storage without affecting schema
B) Changing conceptual schema without affecting applications
C) Changing hardware
D) Changing operating system
ANSWER: B

Q10. Physical data independence refers to:
A) Changing storage structure without affecting conceptual schema
B) Changing tables
C) Changing queries
D) Changing users
ANSWER: A

Q11. A data model describes:
A) Database hardware
B) Structure and relationships of data
C) Network configuration
D) Programming syntax
ANSWER: B

Q12. Which is the most widely used data model?
A) Hierarchical
B) Network
C) Relational
D) Object
ANSWER: C

Q13. In the ER model, a rectangle represents:
A) Attribute
B) Entity
C) Relationship
D) Key
ANSWER: B

Q14. In ER diagrams, an oval represents:
A) Attribute
B) Entity
C) Relationship
D) Key
ANSWER: A

Q15. A diamond symbol in ER diagrams represents:
A) Attribute
B) Entity
C) Relationship
D) Primary key
ANSWER: C

Q16. A table in a relational database is called:
A) Tuple
B) Relation
C) Attribute
D) Schema
ANSWER: B

Q17. A row in a table is known as:
A) Attribute
B) Domain
C) Tuple
D) Field
ANSWER: C

Q18. A column in a table is called:
A) Attribute
B) Relation
C) Tuple
D) Record
ANSWER: A

Q19. A primary key must be:
A) Unique and not null
B) Null
C) Repeating
D) Optional
ANSWER: A

Q20. A foreign key is used to:
A) Delete tables
B) Link two tables
C) Store indexes
D) Store views
ANSWER: B

Q21. Which relational algebra operator selects rows?
A) Projection
B) Selection
C) Join
D) Union
ANSWER: B

Q22. Projection operator is used to:
A) Select rows
B) Select columns
C) Combine tables
D) Delete records
ANSWER: B

Q23. Union operation requires:
A) Same attributes and domains
B) Different attributes
C) Same number of records
D) Same key values
ANSWER: A

Q24. Cartesian product results in:
A) Combination of every row of two tables
B) Only matching rows
C) Only unique rows
D) Only first table rows
ANSWER: A

Q25. The join operation is used to:
A) Remove duplicates
B) Combine related tuples from different relations
C) Delete rows
D) Create indexes
ANSWER: B

Q26. SQL stands for:
A) Structured Query Language
B) Standard Question Language
C) Sequential Query Language
D) System Query Language
ANSWER: A

Q27. Which SQL command retrieves data?
A) INSERT
B) SELECT
C) DELETE
D) UPDATE
ANSWER: B

Q28. Which SQL clause filters rows?
A) WHERE
B) ORDER BY
C) GROUP BY
D) HAVING
ANSWER: A

Q29. Which SQL command adds a new record?
A) UPDATE
B) INSERT
C) SELECT
D) CREATE
ANSWER: B

Q30. Which clause sorts query results?
A) GROUP BY
B) ORDER BY
C) HAVING
D) WHERE
ANSWER: B

Q31. Which SQL statement modifies existing records?
A) UPDATE
B) INSERT
C) DELETE
D) SELECT
ANSWER: A

Q32. A subquery is:
A) Query inside another query
B) Multiple tables
C) Stored procedure
D) Index
ANSWER: A

Q33. Aggregate functions operate on:
A) Multiple rows
B) Single rows
C) Indexes
D) Tables only
ANSWER: A

Q34. Which function returns number of rows?
A) COUNT()
B) SUM()
C) AVG()
D) MAX()
ANSWER: A

Q35. Which clause filters groups?
A) WHERE
B) HAVING
C) ORDER BY
D) SELECT
ANSWER: B

Q36. Normalization is used to:
A) Increase redundancy
B) Reduce redundancy
C) Delete tables
D) Create queries
ANSWER: B

Q37. First Normal Form (1NF) requires:
A) Atomic attributes
B) Multiple values
C) Duplicate records
D) Composite keys only
ANSWER: A

Q38. Second Normal Form requires:
A) 1NF + no partial dependency
B) 1NF only
C) Transitive dependency
D) Duplicate attributes
ANSWER: A

Q39. Third Normal Form removes:
A) Partial dependency
B) Transitive dependency
C) Atomic attributes
D) Keys
ANSWER: B

Q40. Functional dependency means:
A) One attribute determines another
B) Attributes are unrelated
C) Tables are deleted
D) Data is redundant
ANSWER: A

Q41. A transaction is:
A) A single SQL statement
B) A logical unit of database work
C) Hardware operation
D) Query optimization
ANSWER: B

Q42. ACID properties ensure:
A) Data reliability
B) Faster queries
C) Smaller tables
D) Data duplication
ANSWER: A

Q43. Atomicity means:
A) Transaction executed fully or not at all
B) Data stored permanently
C) Data shared among users
D) Queries optimized
ANSWER: A

Q44. Consistency ensures:
A) Database remains valid before and after transaction
B) Queries are fast
C) Tables are indexed
D) Data duplicated
ANSWER: A

Q45. Isolation ensures:
A) Transactions executed independently
B) Data duplication
C) Data deletion
D) Query merging
ANSWER: A

Q46. Concurrency control ensures:
A) Data consistency during simultaneous transactions
B) Faster hardware
C) More storage
D) Data deletion
ANSWER: A

Q47. Deadlock occurs when:
A) Transactions wait indefinitely for resources
B) Queries are slow
C) Tables are deleted
D) Index fails
ANSWER: A

Q48. Two-phase locking protocol ensures:
A) Serializability
B) Redundancy
C) Indexing
D) Data mining
ANSWER: A

Q49. Database recovery is used after:
A) System failure
B) Query execution
C) Table creation
D) Data insertion
ANSWER: A

Q50. The log file stores:
A) Transaction history for recovery
B) User passwords
C) Table names
D) Index values
ANSWER: A

Q51. Which problem occurs when two transactions read and write the same data simultaneously?
A) Deadlock
B) Lost update
C) Data mining
D) Index collision
ANSWER: B

Q52. Dirty read occurs when:
A) A transaction reads uncommitted data
B) Data is deleted
C) Index is updated
D) Query fails
ANSWER: A

Q53. A schedule that produces the same result as serial execution is called:
A) Parallel schedule
B) Serializable schedule
C) Random schedule
D) Non-deterministic schedule
ANSWER: B

Q54. Locking is used in DBMS to:
A) Control access to data
B) Delete tables
C) Store indexes
D) Reduce storage
ANSWER: A

Q55. Shared lock allows:
A) Multiple reads
B) Multiple writes
C) Delete operations
D) Table creation
ANSWER: A

Q56. Which command permanently saves changes?
A) ROLLBACK
B) COMMIT
C) DELETE
D) DROP
ANSWER: B

Q57. ROLLBACK is used to:
A) Save changes
B) Undo changes
C) Create tables
D) Update indexes
ANSWER: B

Q58. Checkpointing helps in:
A) Faster recovery
B) Query optimization
C) Index creation
D) Data mining
ANSWER: A

Q59. Which failure occurs due to power outage?
A) Transaction failure
B) System crash
C) Media failure
D) Logical failure
ANSWER: B

Q60. Media failure refers to:
A) Hard disk crash
B) Query error
C) Syntax error
D) Lock conflict
ANSWER: A

Q61. An index improves:
A) Query speed
B) Storage capacity
C) Data redundancy
D) Data deletion
ANSWER: A

Q62. Which indexing structure is widely used in databases?
A) Binary tree
B) B+ tree
C) Graph
D) Stack
ANSWER: B

Q63. Primary index is built on:
A) Primary key
B) Foreign key
C) Any attribute
D) Non-key attribute
ANSWER: A

Q64. Dense index means:
A) Every search key has an index entry
B) Only some keys have entries
C) Only primary keys indexed
D) Only foreign keys indexed
ANSWER: A

Q65. Sparse index means:
A) Index entry for every record
B) Index entry for some records
C) Index entry for all attributes
D) No index entries
ANSWER: B

Q66. Query processing converts:
A) SQL into execution plan
B) Data into tables
C) Tables into indexes
D) Records into queries
ANSWER: A

Q67. Query optimizer selects:
A) Best execution plan
B) Largest table
C) Smallest database
D) Most queries
ANSWER: A

Q68. Cost-based optimization uses:
A) Estimated cost of operations
B) Table size only
C) User preference
D) Network speed
ANSWER: A

Q69. Heuristic optimization uses:
A) Rules of thumb
B) Random selection
C) User queries
D) Machine learning
ANSWER: A

Q70. Join ordering significantly affects:
A) Query performance
B) Data redundancy
C) Security
D) Storage size
ANSWER: A

Q71. Database security protects:
A) Data from unauthorized access
B) Hardware
C) Operating system
D) Queries
ANSWER: A

Q72. Authentication verifies:
A) User identity
B) Query speed
C) Table design
D) Index type
ANSWER: A

Q73. Authorization determines:
A) What actions a user can perform
B) Query syntax
C) Data storage
D) Table size
ANSWER: A

Q74. SQL command used to give privileges:
A) GRANT
B) REVOKE
C) DELETE
D) UPDATE
ANSWER: A

Q75. SQL command used to remove privileges:
A) REVOKE
B) GRANT
C) INSERT
D) SELECT
ANSWER: A

Q76. Distributed database stores data across:
A) Multiple locations
B) Single computer
C) One table
D) One disk
ANSWER: A

Q77. Data fragmentation means:
A) Dividing database into pieces
B) Deleting tables
C) Merging tables
D) Compressing data
ANSWER: A

Q78. Replication means:
A) Copying data across sites
B) Deleting duplicates
C) Query processing
D) Indexing
ANSWER: A

Q79. Transparency in distributed DBMS means:
A) Users unaware of distribution
B) Visible network
C) Visible queries
D) Visible schema
ANSWER: A

Q80. Which protocol ensures distributed transaction consistency?
A) Two-phase commit
B) Three-level schema
C) Two-phase locking
D) Data mining protocol
ANSWER: A

Q81. NoSQL databases are designed for:
A) Large-scale distributed data
B) Small local files
C) Spreadsheet data
D) Text editing
ANSWER: A

Q82. Which is a document-based NoSQL database?
A) MongoDB
B) MySQL
C) Oracle
D) PostgreSQL
ANSWER: A

Q83. Key-value databases store:
A) Pairs of keys and values
B) Tables
C) Columns
D) Rows
ANSWER: A

Q84. Which NoSQL type stores graph relationships?
A) Graph databases
B) Column databases
C) Relational databases
D) File databases
ANSWER: A

Q85. Cassandra is an example of:
A) Column-oriented database
B) Relational database
C) Hierarchical database
D) Network database
ANSWER: A

Q86. A data warehouse is used for:
A) Analytical processing
B) Transaction processing
C) Data deletion
D) Query optimization
ANSWER: A

Q87. OLAP stands for:
A) Online Analytical Processing
B) Online Application Processing
C) Object Level Access Protocol
D) Open Logical Access Protocol
ANSWER: A

Q88. OLTP systems handle:
A) Day-to-day transactions
B) Data mining
C) Data warehousing
D) Reporting only
ANSWER: A

Q89. Data mining is used to:
A) Discover patterns in data
B) Delete data
C) Create indexes
D) Encrypt data
ANSWER: A

Q90. Which technique groups similar data?
A) Clustering
B) Sorting
C) Joining
D) Indexing
ANSWER: A

Q91. If relation R has 5 tuples and S has 4 tuples, the Cartesian product will have:
A) 9 tuples
B) 20 tuples
C) 5 tuples
D) 4 tuples
ANSWER: B

Q92. Which normal form eliminates transitive dependency?
A) 1NF
B) 2NF
C) 3NF
D) BCNF
ANSWER: C

Q93. BCNF is stronger than:
A) 1NF
B) 2NF
C) 3NF
D) All of the above
ANSWER: D

Q94. Which SQL clause is executed first logically?
A) SELECT
B) FROM
C) WHERE
D) ORDER BY
ANSWER: B

Q95. Which anomaly occurs when inserting data requires unrelated information?
A) Insertion anomaly
B) Update anomaly
C) Deletion anomaly
D) Join anomaly
ANSWER: A

Q96. Which anomaly causes unintended data loss?
A) Deletion anomaly
B) Insertion anomaly
C) Update anomaly
D) Lock anomaly
ANSWER: A

Q97. A candidate key is:
A) Minimal super key
B) Any key
C) Foreign key
D) Secondary key
ANSWER: A

Q98. A super key is:
A) Attribute set uniquely identifying tuples
B) Foreign key
C) Composite key only
D) Index key
ANSWER: A

Q99. In B+ trees, data pointers are stored in:
A) Leaf nodes
B) Root nodes
C) Internal nodes
D) All nodes
ANSWER: A

Q100. Hash indexing is efficient for:
A) Equality search
B) Range queries
C) Join queries
D) Sorting
ANSWER: A

Q101. Which SQL command removes a table?
A) DELETE
B) DROP
C) REMOVE
D) CLEAR
ANSWER: B

Q102. Which SQL command removes rows but keeps table structure?
A) DROP
B) DELETE
C) REMOVE
D) ERASE
ANSWER: B

Q103. View in SQL is:
A) Virtual table
B) Physical table
C) Index
D) File
ANSWER: A

Q104. Stored procedures are used for:
A) Reusable SQL logic
B) Indexing
C) Storage
D) Encryption
ANSWER: A

Q105. Trigger executes:
A) Automatically on events
B) Manually only
C) During indexing
D) During backup
ANSWER: A

Q106. Which join returns matching rows only?
A) Inner join
B) Left join
C) Right join
D) Full join
ANSWER: A

Q107. LEFT JOIN returns:
A) All rows from left table
B) All rows from right table
C) Only matching rows
D) No rows
ANSWER: A

Q108. RIGHT JOIN returns:
A) All rows from right table
B) All rows from left table
C) Only matching rows
D) No rows
ANSWER: A

Q109. FULL JOIN returns:
A) All rows from both tables
B) Only matching rows
C) Only left table rows
D) Only right table rows
ANSWER: A

Q110. Self join joins:
A) Table with itself
B) Two databases
C) Two schemas
D) Two indexes
ANSWER: A

Q111. Which constraint prevents duplicate values?
A) UNIQUE
B) NOT NULL
C) CHECK
D) DEFAULT
ANSWER: A

Q112. NOT NULL ensures:
A) Attribute cannot be empty
B) Attribute unique
C) Attribute indexed
D) Attribute encrypted
ANSWER: A

Q113. CHECK constraint enforces:
A) Specific condition
B) Indexing
C) Security
D) Query speed
ANSWER: A

Q114. DEFAULT constraint provides:
A) Automatic value
B) Key value
C) Index value
D) Foreign value
ANSWER: A

Q115. Data dictionary stores:
A) Metadata
B) Data records
C) Queries
D) Indexes
ANSWER: A

Q116. Metadata means:
A) Data about data
B) Raw data
C) Duplicate data
D) Temporary data
ANSWER: A

Q117. Which language defines database structure?
A) DDL
B) DML
C) DCL
D) TCL
ANSWER: A

Q118. Which language manipulates data?
A) DML
B) DDL
C) DCL
D) TCL
ANSWER: A

Q119. Which language controls permissions?
A) DCL
B) DDL
C) DML
D) TCL
ANSWER: A

Q120. Which command saves transaction changes?
A) COMMIT
B) ROLLBACK
C) DELETE
D) UPDATE
ANSWER: A

Q121. Which SQL clause groups rows with similar values?
A) ORDER BY
B) GROUP BY
C) WHERE
D) SELECT
ANSWER: B

Q122. Which clause filters groups created by GROUP BY?
A) WHERE
B) HAVING
C) ORDER BY
D) DISTINCT
ANSWER: B

Q123. Which keyword removes duplicate rows in SQL results?
A) UNIQUE
B) DISTINCT
C) GROUP
D) FILTER
ANSWER: B

Q124. Which command creates a new table?
A) CREATE TABLE
B) NEW TABLE
C) ADD TABLE
D) MAKE TABLE
ANSWER: A

Q125. Which command changes table structure?
A) MODIFY TABLE
B) ALTER TABLE
C) CHANGE TABLE
D) UPDATE TABLE
ANSWER: B

Q126. A composite key consists of:
A) Single attribute
B) Multiple attributes
C) Only primary key
D) Only foreign key
ANSWER: B

Q127. Which key references a primary key in another table?
A) Candidate key
B) Foreign key
C) Composite key
D) Super key
ANSWER: B

Q128. Referential integrity ensures:
A) Foreign key values must match primary key values
B) Tables are sorted
C) Queries are fast
D) Data is encrypted
ANSWER: A

Q129. A candidate key that is chosen to identify tuples is:
A) Alternate key
B) Primary key
C) Foreign key
D) Composite key
ANSWER: B

Q130. Candidate keys not selected as primary key are called:
A) Foreign keys
B) Alternate keys
C) Secondary keys
D) Index keys
ANSWER: B

Q131. Intersection operation returns:
A) Common tuples from relations
B) All tuples
C) Different tuples
D) Cartesian tuples
ANSWER: A

Q132. Difference operation returns:
A) Tuples present in first relation but not second
B) All tuples
C) Common tuples
D) Combined tuples
ANSWER: A

Q133. Natural join combines relations based on:
A) Same attribute names
B) Different attributes
C) Random attributes
D) All attributes
ANSWER: A

Q134. Division operation is useful for:
A) Queries involving "for all" conditions
B) Sorting data
C) Indexing
D) Updating records
ANSWER: A

Q135. Rename operator in relational algebra is:
A) ρ (rho)
B) σ (sigma)
C) π (pi)
D) ∪ (union)
ANSWER: A

Q136. Partial dependency occurs when:
A) Non-key depends on part of composite key
B) Non-key depends on full key
C) Key depends on non-key
D) Attribute independent
ANSWER: A

Q137. Transitive dependency occurs when:
A) A→B and B→C
B) A→C directly
C) A→A
D) C→A
ANSWER: A

Q138. Boyce–Codd Normal Form requires:
A) Every determinant is a candidate key
B) No partial dependency
C) No transitive dependency
D) Atomic attributes
ANSWER: A

Q139. Denormalization is used to:
A) Improve performance
B) Reduce tables
C) Increase redundancy intentionally
D) Both A and C
ANSWER: D

Q140. Functional dependency notation is:
A) A → B
B) A = B
C) A <> B
D) A || B
ANSWER: A

Q141. Durability means:
A) Changes survive system failure
B) Transactions execute once
C) Data remains hidden
D) Tables locked
ANSWER: A

Q142. Which property prevents interference between transactions?
A) Atomicity
B) Isolation
C) Consistency
D) Durability
ANSWER: B

Q143. Write-Ahead Logging requires:
A) Log written before database update
B) Database written first
C) Both simultaneously
D) No log required
ANSWER: A

Q144. Cascading rollback occurs when:
A) One rollback triggers others
B) Transactions commit together
C) Data deleted
D) Locks removed
ANSWER: A

Q145. A schedule without cascading rollback is called:
A) Recoverable schedule
B) Cascadeless schedule
C) Serial schedule
D) Parallel schedule
ANSWER: B

Q146. Two-phase locking has:
A) Growing and shrinking phases
B) Reading and writing phases
C) Lock and unlock phases
D) Parallel phases
ANSWER: A

Q147. Lock conversion allows:
A) Changing shared lock to exclusive lock
B) Deleting lock
C) Removing data
D) Creating indexes
ANSWER: A

Q148. Starvation occurs when:
A) Transaction waits indefinitely
B) Data lost
C) Table locked permanently
D) Disk failure
ANSWER: A

Q149. Timestamp ordering uses:
A) Transaction timestamps
B) Lock tables
C) Random scheduling
D) Priority scheduling
ANSWER: A

Q150. Optimistic concurrency control assumes:
A) Conflicts are rare
B) Conflicts frequent
C) No transactions
D) Locks always required
ANSWER: A

Q151. Secondary index is created on:
A) Non-key attribute
B) Primary key
C) Foreign key only
D) Composite key only
ANSWER: A

Q152. B+ tree height affects:
A) Search time
B) Storage
C) Data redundancy
D) Security
ANSWER: A

Q153. Hash table collision occurs when:
A) Two keys map to same location
B) No key found
C) Key deleted
D) Index removed
ANSWER: A

Q154. Clustered index determines:
A) Physical order of data
B) Logical order
C) Query speed only
D) Security
ANSWER: A

Q155. Non-clustered index stores:
A) Pointers to data rows
B) Actual data rows
C) Table structure
D) Queries
ANSWER: A

Q156. Query plan is:
A) Execution strategy
B) SQL query
C) Database schema
D) Table structure
ANSWER: A

Q157. Selection pushdown means:
A) Apply selection early
B) Apply selection last
C) Remove selection
D) Duplicate selection
ANSWER: A

Q158. Join algorithms include:
A) Nested loop join
B) Merge join
C) Hash join
D) All of the above
ANSWER: D

Q159. Which join algorithm works well for sorted relations?
A) Merge join
B) Nested join
C) Hash join
D) Cross join
ANSWER: A

Q160. Which join algorithm uses hashing?
A) Hash join
B) Merge join
C) Loop join
D) Self join
ANSWER: A

Q161. Encryption protects:
A) Data confidentiality
B) Query speed
C) Indexing
D) Table creation
ANSWER: A

Q162. Role-based access control assigns permissions to:
A) Roles
B) Users directly
C) Queries
D) Tables
ANSWER: A

Q163. SQL injection is:
A) Security attack through queries
B) Query optimization
C) Index error
D) Data mining
ANSWER: A

Q164. Audit logs track:
A) Database activities
B) Queries only
C) Tables only
D) Index changes
ANSWER: A

Q165. Data masking is used to:
A) Hide sensitive data
B) Encrypt tables
C) Delete records
D) Speed queries
ANSWER: A

Q166. Horizontal fragmentation divides table by:
A) Rows
B) Columns
C) Keys
D) Indexes
ANSWER: A

Q167. Vertical fragmentation divides table by:
A) Columns
B) Rows
C) Indexes
D) Keys
ANSWER: A

Q168. Replication improves:
A) Availability
B) Redundancy control
C) Security
D) Query syntax
ANSWER: A

Q169. Distributed query processing occurs when:
A) Query executed across multiple sites
B) Single site processing
C) Data deleted
D) Query optimized
ANSWER: A

Q170. Data localization converts:
A) Global query into local queries
B) SQL into algebra
C) Tables into indexes
D) Data into queries
ANSWER: A

Q171. CAP theorem includes:
A) Consistency
B) Availability
C) Partition tolerance
D) All of the above
ANSWER: D

Q172. Document databases store data in:
A) JSON-like documents
B) Tables
C) Graphs
D) Arrays
ANSWER: A

Q173. Graph databases are ideal for:
A) Relationship-heavy data
B) Numeric data
C) Logs
D) Images
ANSWER: A

Q174. Which database is graph-based?
A) Neo4j
B) MySQL
C) Oracle
D) SQL Server
ANSWER: A

Q175. Big data databases prioritize:
A) Scalability
B) Small storage
C) Limited queries
D) Simple tables
ANSWER: A

Q176. ETL stands for:
A) Extract Transform Load
B) Execute Transfer Load
C) Extract Transfer Logic
D) Execute Transform Logic
ANSWER: A

Q177. Fact table stores:
A) Quantitative data
B) Dimension names
C) Metadata
D) Index values
ANSWER: A

Q178. Dimension tables store:
A) Descriptive attributes
B) Numeric measures
C) Logs
D) Keys only
ANSWER: A

Q179. Star schema consists of:
A) One fact table and multiple dimensions
B) Multiple fact tables
C) Only dimension tables
D) Only fact tables
ANSWER: A

Q180. Snowflake schema is:
A) Normalized dimension tables
B) Single dimension table
C) Only fact tables
D) Random tables
ANSWER: A

Q181. If relation has attributes A,B,C and functional dependency A→B, then A is:
A) Determinant
B) Dependent
C) Foreign key
D) Index key
ANSWER: A

Q182. If R(A,B,C) and A→B and B→C then A→C due to:
A) Transitivity
B) Reflexivity
C) Augmentation
D) Projection
ANSWER: A

Q183. Which property states if Y⊆X then X→Y?
A) Reflexivity
B) Transitivity
C) Augmentation
D) Decomposition
ANSWER: A

Q184. Which rule states if X→Y then XZ→YZ?
A) Augmentation
B) Transitivity
C) Reflexivity
D) Union
ANSWER: A

Q185. Decomposition rule means:
A) X→YZ implies X→Y and X→Z
B) X→Y implies Y→X
C) X→Z implies Y→Z
D) X→X
ANSWER: A

Q186. Database schema describes:
A) Structure of database
B) Data values
C) Queries
D) Indexes
ANSWER: A

Q187. Instance refers to:
A) Data stored at specific time
B) Table structure
C) Query structure
D) Schema definition
ANSWER: A

Q188. Logical schema defines:
A) Tables and relationships
B) Physical storage
C) Hardware
D) Operating system
ANSWER: A

Q189. Physical schema defines:
A) Storage structure
B) Queries
C) Tables only
D) Keys only
ANSWER: A

Q190. Data abstraction levels include:
A) Physical
B) Logical
C) View
D) All of the above
ANSWER: D

Q191. A view hides:
A) Complexity of database
B) Data
C) Queries
D) Indexes
ANSWER: A

Q192. Which SQL command deletes all rows but keeps structure?
A) TRUNCATE
B) DELETE
C) DROP
D) REMOVE
ANSWER: A

Q193. TRUNCATE is faster than DELETE because:
A) It does not log individual rows
B) It deletes indexes
C) It deletes tables
D) It removes schema
ANSWER: A

Q194. Which command removes database entirely?
A) DROP DATABASE
B) DELETE DATABASE
C) REMOVE DATABASE
D) CLEAR DATABASE
ANSWER: A

Q195. Materialized view stores:
A) Actual query results
B) Virtual results
C) Temporary queries
D) Index values
ANSWER: A

Q196. Which SQL function finds maximum value?
A) MAX()
B) SUM()
C) AVG()
D) COUNT()
ANSWER: A

Q197. Which function calculates average?
A) AVG()
B) SUM()
C) COUNT()
D) MAX()
ANSWER: A

Q198. Which SQL operator tests membership?
A) IN
B) BETWEEN
C) LIKE
D) EXISTS
ANSWER: A

Q199. LIKE operator is used for:
A) Pattern matching
B) Equality test
C) Range test
D) Join operation
ANSWER: A

Q200. BETWEEN operator checks:
A) Range of values
B) Pattern match
C) Equality
D) Join condition
ANSWER: A

Q201. Which SQL clause is used to rename a column in the output?
A) AS
B) RENAME
C) CHANGE
D) MODIFY
ANSWER: A

Q202. Which SQL operator checks whether a subquery returns any row?
A) EXISTS
B) IN
C) LIKE
D) ANY
ANSWER: A

Q203. Which operator compares a value to any value returned by a subquery?
A) ALL
B) ANY
C) EXISTS
D) IN
ANSWER: B

Q204. Which SQL keyword ensures only unique values are inserted?
A) UNIQUE
B) DISTINCT
C) PRIMARY
D) CHECK
ANSWER: A

Q205. A correlated subquery:
A) Executes once
B) Executes for each row of outer query
C) Executes only after join
D) Executes before outer query
ANSWER: B

Q206. Parsing in query processing checks:
A) Syntax and structure
B) Data values
C) Indexing
D) Storage
ANSWER: A

Q207. Query rewriting transforms:
A) SQL into optimized form
B) SQL into table
C) Data into schema
D) Index into table
ANSWER: A

Q208. Execution engine performs:
A) Actual query execution
B) Query design
C) Table design
D) Data entry
ANSWER: A

Q209. Selectivity of a condition refers to:
A) Fraction of rows satisfying condition
B) Number of tables
C) Number of queries
D) Number of joins
ANSWER: A

Q210. Cost estimation depends mainly on:
A) Disk I/O
B) Table names
C) SQL syntax
D) User input
ANSWER: A

Q211. Heap file organization stores records in:
A) Random order
B) Sorted order
C) Indexed order
D) Clustered order
ANSWER: A

Q212. Sequential file organization stores records:
A) Sorted by key
B) Randomly
C) By hash
D) By index pointer
ANSWER: A

Q213. Overflow blocks occur in:
A) Hash indexing
B) Sequential indexing
C) B+ trees
D) Clustered indexing
ANSWER: A

Q214. Multilevel indexing reduces:
A) Search time
B) Storage
C) Security
D) Redundancy
ANSWER: A

Q215. In B+ trees, internal nodes store:
A) Search keys
B) Data records
C) Entire tables
D) Metadata
ANSWER: A

Q216. Which mechanism prevents unauthorized data modification?
A) Access control
B) Backup
C) Indexing
D) Data mining
ANSWER: A

Q217. Data encryption converts:
A) Plain text into cipher text
B) Data into tables
C) Tables into indexes
D) Queries into data
ANSWER: A

Q218. Which security mechanism tracks user actions?
A) Auditing
B) Indexing
C) Hashing
D) Normalization
ANSWER: A

Q219. Least privilege principle means:
A) Users get minimum permissions required
B) Users get full access
C) Users get read-only access
D) Users get admin access
ANSWER: A

Q220. Which attack manipulates SQL queries via user input?
A) SQL injection
B) Phishing
C) Malware
D) Spoofing
ANSWER: A

Q221. Which type of fragmentation replicates full tables?
A) Replication
B) Horizontal fragmentation
C) Vertical fragmentation
D) Hybrid fragmentation
ANSWER: A

Q222. Distributed DBMS improves:
A) Availability and reliability
B) Query complexity
C) Data redundancy
D) Hardware cost
ANSWER: A

Q223. Distributed transactions require:
A) Coordination among sites
B) Local queries only
C) Single server
D) Single table
ANSWER: A

Q224. Two-phase commit protocol ensures:
A) Atomic distributed transactions
B) Faster queries
C) Data mining
D) Indexing
ANSWER: A

Q225. Coordinator in distributed commit:
A) Manages commit decision
B) Executes query
C) Stores data
D) Creates index
ANSWER: A

Q226. Column-family databases store data by:
A) Columns
B) Rows
C) Documents
D) Graphs
ANSWER: A

Q227. Which NoSQL database uses key-value storage?
A) Redis
B) MySQL
C) Oracle
D) PostgreSQL
ANSWER: A

Q228. BASE properties in NoSQL mean:
A) Basically Available Soft-state Eventually consistent
B) Binary Access Storage Engine
C) Basic Application System Engine
D) Base Access Storage Element
ANSWER: A

Q229. Eventual consistency means:
A) Data will become consistent over time
B) Data always consistent instantly
C) Data never consistent
D) Data deleted automatically
ANSWER: A

Q230. NoSQL databases are commonly used in:
A) Big data applications
B) Small spreadsheets
C) Local text files
D) Single-user programs
ANSWER: A

Q231. Data warehouse integrates data from:
A) Multiple sources
B) Single table
C) Single application
D) Single database
ANSWER: A

Q232. Data warehouse data is typically:
A) Historical
B) Temporary
C) Random
D) Unstructured only
ANSWER: A

Q233. OLAP cube represents:
A) Multidimensional data
B) Single dimension
C) Binary data
D) Text data
ANSWER: A

Q234. Drill-down operation:
A) Shows more detailed data
B) Shows summarized data
C) Deletes data
D) Filters rows
ANSWER: A

Q235. Roll-up operation:
A) Aggregates data
B) Deletes rows
C) Splits data
D) Updates data
ANSWER: A

Q236. Classification predicts:
A) Category labels
B) Numeric values
C) Tables
D) Queries
ANSWER: A

Q237. Regression predicts:
A) Numeric values
B) Categories
C) Tables
D) Keys
ANSWER: A

Q238. Association rule mining finds:
A) Relationships between items
B) Data deletion rules
C) Query optimization
D) Index rules
ANSWER: A

Q239. Support in association rules measures:
A) Frequency of itemset
B) Query cost
C) Index size
D) Data redundancy
ANSWER: A

Q240. Confidence in association rules measures:
A) Rule reliability
B) Query time
C) Data size
D) Storage cost
ANSWER: A

Q241. If relation R has 6 tuples and S has 3 tuples, their Cartesian product has:
A) 9 tuples
B) 18 tuples
C) 6 tuples
D) 3 tuples
ANSWER: B

Q242. A relation is in BCNF if:
A) Every determinant is a candidate key
B) Only primary keys exist
C) Only foreign keys exist
D) Only composite keys exist
ANSWER: A

Q243. If a table has no partial dependency but has transitive dependency, it is in:
A) 2NF
B) 1NF
C) 3NF
D) BCNF
ANSWER: A

Q244. A relation in 3NF may still violate:
A) BCNF
B) 2NF
C) 1NF
D) None
ANSWER: A

Q245. If $X \rightarrow Y$ and $Y \rightarrow Z$ then $X \rightarrow Z$ by:
A) Transitivity
B) Reflexivity
C) Augmentation
D) Union
ANSWER: A

Q246. Which SQL clause eliminates duplicates in GROUP results?
A) DISTINCT
B) UNIQUE
C) GROUP BY
D) HAVING
ANSWER: A

Q247. Which SQL command updates multiple rows?
A) UPDATE
B) MODIFY
C) CHANGE
D) INSERT
ANSWER: A

Q248. Which join returns rows with no matching rows in another table?
A) Outer join
B) Inner join
C) Cross join
D) Natural join
ANSWER: A

Q249. Which SQL keyword limits result rows?
A) LIMIT
B) RANGE
C) TOP
D) STOP
ANSWER: A

Q250. Which SQL operator matches wildcard patterns?
A) LIKE
B) MATCH
C) EQUAL
D) IN
ANSWER: A

Q251. Redundancy means:
A) Duplicate data
B) Unique data
C) Secure data
D) Encrypted data
ANSWER: A

Q252. Data inconsistency occurs when:
A) Same data differs in multiple places
B) Data duplicated
C) Data indexed
D) Data encrypted
ANSWER: A

Q253. Data independence allows:
A) Changing schema without affecting programs
B) Deleting tables
C) Adding indexes
D) Changing hardware
ANSWER: A

Q254. DBMS replaces:
A) File-based systems
B) Operating systems
C) Compilers
D) Browsers
ANSWER: A

Q255. Data abstraction hides:
A) Complex storage details
B) Queries
C) Tables
D) Keys
ANSWER: A

Q256. If a table has 4 attributes, the maximum number of candidate keys depends on:
A) Functional dependencies
B) Number of rows
C) Indexes
D) Queries
ANSWER: A

Q257. If a transaction fails before commit:
A) It is rolled back
B) It is saved
C) It is locked
D) It is indexed
ANSWER: A

Q258. Which schedule guarantees serializability?
A) Conflict-serializable schedule
B) Random schedule
C) Parallel schedule
D) Dynamic schedule
ANSWER: A

Q259. Which concurrency problem causes inconsistent updates?
A) Lost update
B) Dirty read
C) Phantom read
D) Deadlock
ANSWER: A

Q260. Phantom read occurs when:
A) New rows appear in repeated query
B) Data deleted
C) Data duplicated
D) Index corrupted
ANSWER: A

Q261. Deadlock detection requires:
A) Wait-for graph
B) Query plan
C) Data mining
D) Index tree
ANSWER: A

Q262. Deadlock prevention uses:
A) Timestamp ordering
B) Data mining
C) Query parsing
D) Data indexing
ANSWER: A

Q263. Which recovery technique replays log entries?
A) Redo
B) Undo
C) Delete
D) Drop
ANSWER: A

Q264. Undo operation reverses:
A) Uncommitted transactions
B) Committed transactions
C) Queries
D) Indexing
ANSWER: A

Q265. Checkpoints reduce:
A) Recovery time
B) Storage
C) Query speed
D) Table size
ANSWER: A

Q266. Bitmap index works best for:
A) Low-cardinality attributes
B) High-cardinality attributes
C) Large tables
D) Numeric keys
ANSWER: A

Q267. Hashing is inefficient for:
A) Range queries
B) Equality search
C) Key lookup
D) Index search
ANSWER: A

Q268. Clustered index allows:
A) Faster range queries
B) Slower queries
C) Data duplication
D) Query deletion
ANSWER: A

Q269. Query optimizer mainly aims to:
A) Minimize execution cost
B) Increase queries
C) Increase tables
D) Increase joins
ANSWER: A

Q270. Query cost is mostly influenced by:
A) Disk access
B) CPU clock
C) Monitor size
D) Network cable
ANSWER: A

Q271. Data warehouse supports:
A) Decision support systems
B) Transaction processing
C) Data entry
D) Query deletion
ANSWER: A

Q272. Data mart is:
A) Subset of data warehouse
B) Entire warehouse
C) Database server
D) Query optimizer
ANSWER: A

Q273. Data cube enables:
A) Multidimensional analysis
B) Table deletion
C) Query optimization
D) Indexing
ANSWER: A

Q274. Slice operation selects:
A) Single dimension value
B) Entire cube
C) Multiple cubes
D) Table structure
ANSWER: A

Q275. Dice operation selects:
A) Sub-cube
B) Entire cube
C) Table
D) Index
ANSWER: A

Q276. Data mining helps in:
A) Knowledge discovery
B) Query optimization
C) Table creation
D) Indexing
ANSWER: A

Q277. Clustering groups:
A) Similar data points
B) Different tables
C) Queries
D) Keys
ANSWER: A

Q278. Outlier detection identifies:
A) Abnormal data patterns
B) Tables
C) Indexes
D) Queries
ANSWER: A

Q279. Decision trees are used for:
A) Classification
B) Indexing
C) Query planning
D) Recovery
ANSWER: A

Q280. Neural networks are used in:
A) Predictive data mining
B) Indexing
C) Transactions
D) Normalization
ANSWER: A

Q281. OLTP systems focus on:
A) Fast transaction processing
B) Data mining
C) Historical analysis
D) Reporting
ANSWER: A

Q282. OLAP systems focus on:
A) Data analysis
B) Transactions
C) Data entry
D) Index creation
ANSWER: A

Q283. Metadata describes:
A) Structure of data
B) Actual data
C) Queries
D) Transactions
ANSWER: A

Q284. Data lineage tracks:
A) Data origin and movement
B) Queries
C) Indexes
D) Transactions
ANSWER: A

Q285. Schema evolution allows:
A) Schema modification over time
B) Data deletion
C) Query deletion
D) Index deletion
ANSWER: A

Q286. View-based security restricts:
A) Data visibility
B) Query speed
C) Storage
D) Indexing
ANSWER: A

Q287. Data replication improves:
A) Fault tolerance
B) Query syntax
C) Indexing
D) Table design
ANSWER: A

Q288. Horizontal scaling means:
A) Adding more machines
B) Increasing CPU
C) Increasing RAM
D) Increasing disk size
ANSWER: A

Q289. Vertical scaling means:
A) Increasing machine capacity
B) Adding machines
C) Deleting machines
D) Replicating machines
ANSWER: A

Q290. CAP theorem states distributed systems cannot guarantee simultaneously:
A) Consistency, Availability, Partition tolerance
B) Speed, size, cost
C) Tables, keys, indexes
D) Queries, joins, views
ANSWER: A

Q291. Strong consistency means:
A) All nodes see same data instantly
B) Data eventually consistent
C) Data randomly consistent
D) Data partially consistent
ANSWER: A

Q292. Partition tolerance means:
A) System continues despite network failures
B) System stops completely
C) Data deleted
D) Queries blocked
ANSWER: A

Q293. Big data systems prioritize:
A) Scalability
B) Small storage
C) Single user
D) Simple queries
ANSWER: A

Q294. Hadoop ecosystem supports:
A) Distributed data processing
B) Relational queries
C) Small databases
D) Local storage
ANSWER: A

Q295. MapReduce processes:
A) Large datasets in parallel
B) Small tables
C) SQL queries
D) Index structures
ANSWER: A

Q296. Data lake stores:
A) Raw structured and unstructured data
B) Only tables
C) Only indexes
D) Only queries
ANSWER: A

Q297. Schema-on-read is typical of:
A) Data lakes
B) Data warehouses
C) Relational DBMS
D) OLTP systems
ANSWER: A

Q298. Schema-on-write is typical of:
A) Data warehouses
B) Data lakes
C) NoSQL systems
D) Graph databases
ANSWER: A

Q299. Query federation allows:
A) Querying multiple databases simultaneously
B) Query deletion
C) Data mining
D) Indexing
ANSWER: A

Q300. The main goal of DBMS is to:
A) Efficiently store, retrieve, and manage data
B) Replace operating systems
C) Replace programming languages
D) Eliminate queries
ANSWER: A